



IT-5006 Fundamentals of Data Analytics

Team Assignment Phase One Report

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1. Introduction

This interim report covers the initial part of the data analysis of Chicago crime data.

The team conducted a comprehensive literature review on data-analysis methodologies, including research on existing predictive policing systems and analytical frameworks used in prior studies.

The report also covers the initial stages of data preprocessing, exploratory analysis, and the identification of key variables that will inform the subsequent modelling phase.

The interim findings also guide the design of the interactive dashboard, which enables users to visualize crime patterns across multiple dimensions and supports hypothesis generation for the final analysis.

2. Literature Review

- Summary of existing systems and their effectiveness
- Review of modeling approaches
- Feature engineering techniques
- Evaluation metrics and their appropriateness
- Conclusion: key trends, gaps, and areas of improvement in future

Existing systems and their effectiveness:

Predictive policing systems have evolved from early data-driven policing models to modern algorithmic predictive systems. Early systems, such as CompStat, primarily supported policing decisions and resource allocation by integrating historical crime data, while still retaining officers' judgment in actual decision-making. These systems improved policing management efficiency and promoted the application of data in policing work [1].

With the development of data analytics, modern predictive policing systems are increasingly adopting machine learning algorithms, typified by PredPol, to predict crime hotspots. PredPol predicts areas where crime may occur in the future based on historical crime types, locations, and times. Simultaneously, it helps law enforcement agencies improve patrol strategies and law enforcement efficiency through patrol time monitoring and resource allocation functions [2]. PredPol has been applied by numerous law enforcement agencies worldwide, demonstrating its high operability and widespread value in practical policing environments.

Real-Time Crime Centers (RTCCs) represent another type of integrated technology platform. RTCCs enhance police capabilities in analyzing and investigating crimes by integrating data from various sources and surveillance technologies [3]. Some studies have shown a significant correlation between the establishment of RTCCs and increased crime clearance rates. Clearance rates for various types of crimes have increased to varying degrees. This demonstrates the system's positive role in improving policing efficiency.

However, despite the widespread adoption of predictive policing systems in practice, their effectiveness remains controversial. Some studies suggest that predictive systems are not significantly effective in reducing crime rates or improving policing efficiency [4]. Currently, systematic empirical evidence regarding the effectiveness of predictive policing is still limited, and related policy practices are supported to some extent only by theoretical speculation.

Review of modelling approaches:

In crime prediction research, various modeling methods are widely used to analyze historical crime data and predict future crime trends. Existing research typically categorizes crime prediction models into two main types: statistical models and artificial intelligence models. Statistical models were the primary method in early crime prediction research, with common techniques including linear regression, exponential smoothing, and autoregressive integral moving average (ARIMA) models. These methods perform well when dealing with relatively simple and linearly distributed data but are often limited by the complexity of crime data in practice [5].

With the development of technology, artificial intelligence (AI) technology is increasingly being applied to the field of crime prediction [5]. AI models, such as artificial neural networks and support vector regression, can effectively capture nonlinear relationships in crime data and demonstrate stronger predictive capabilities when dealing with complex environments and large-scale data. Furthermore, some studies have proposed combining statistical models with AI models, integrating linear and nonlinear features to improve prediction accuracy.

In recent years, spatial and spatiotemporal crime prediction models have gradually become a research focus. Crime prediction methods can be further subdivided into kernel density estimation, point process models, traditional machine learning models, and deep learning models [6]. Hotspot prediction is the most common application scenario, and machine learning methods are widely used due to their high predictive performance. Meanwhile, deep learning models have shown great potential in processing complex spatiotemporal crime data.

In general, different models have their own advantages. Statistical models are suitable for data with simpler structures, while artificial intelligence and deep learning models are better suited for handling complex and dynamically changing data. In practical applications, it is usually necessary to comprehensively consider data characteristics, data size, and prediction objectives to select the appropriate model.

Feature engineering techniques:

To improve the prediction and hotspot identification, it is necessary to do Data Cleaning first. This includes transforming the raw time fields into cyclical variables. Also, Geographic data needs to be validated to ensure that only records with precise coordinates or administrative beats are used for spatial analysis.

Next is Feature Construction. This includes Spatial to identify public and private spaces and Statistical(Z-Score) to indicate anomalies.

After all the processes, the data will be put into Feature Selection. This is to remove the

redundancy using Pearson correlation and mutual information, then use model-based importance and cross-validation to select a high-value, stable subset of predictors.

Evaluation metrics and their appropriateness:

Since the crime data is imbalanced, a hybrid evaluation framework is applied.

For classification tasks, Precision, Recall, and the F1-score are applied to better reflect the trade-offs in resource allocation and the public safety costs of missed detections.

Performance in low-prevalence settings is further scrutinized using AUPRC and AUC-ROC. On the regression side, we utilize MAE for its robustness to outliers and RMSE to penalize large fluctuations.

To bridge the gap between theory and practice, we incorporate spatial-specific indicators like the Hit Rate and the Prediction Accuracy Index (PAI), which measure the model's utility within the actual constraints of patrol coverage.

Conclusion:

Current trends in the field show a clear evolution from traditional statistics toward spatiotemporal deep learning, yet several critical gaps persist. Despite the proliferation of real-time predictive dashboards, there is still a lack of long-term empirical evidence confirming their impact on crime reduction, and many models suffer from a "black box" nature or reflect historical biases in policing data.

Furthermore, the limited visibility of incidents in private residences creates a data quality disparity compared to public spaces. Future efforts should prioritize the integration of contextual variables like urban mobility and weather, the adoption of explainable AI (XAI) frameworks for better transparency, and the development of bias-auditing mechanisms to ensure that predictive tools are both ethically sound and operationally effective.

3. Data Analysis

1.

To more accurately identify “abnormal” situations, we have introduced the Z-Score statistic. The Z-Score measures the degree to which a specific crime rate in a police district deviates from the city average. A score greater than 2.0 indicates that the crime pattern in that district exhibits statistically significant abnormality. This approach helps us filter out the “true pain points requiring attention” from the cluttered data

Referencing the heatmap, we identified the following key anomalies: 1. Burglary Anomaly Zone: District 16 achieved a Z-Score of 2.35. This indicates that while the district's overall crime volume may not rank high, its burglary activity exhibits an extremely high degree of “specialization.” This typically indicates potential systemic security vulnerabilities in the area (e.g., specific residential structures). 2. Motor Vehicle Theft Anomaly Zone: District 2's Z-Scores for both motor vehicle theft and criminal damage exceed 2.0. This dual high score suggests an active criminal network in the area, such as unregulated parking lots becoming hotbeds for theft and deliberate destruction. 3.

Violence Core Zone: District 3 exhibits the most pronounced anomaly in ASSAULT scores.

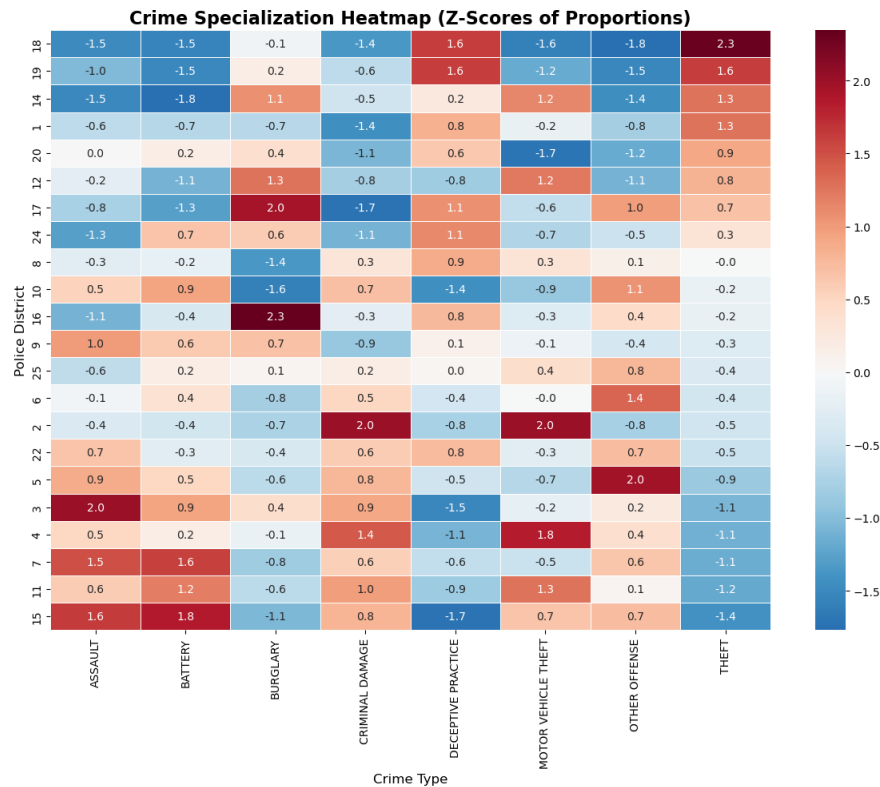


Figure1.district_specialization_heatmap_zscore

2.

Analysis of the dataset reveals a stark correlation between the physical environment and enforcement success, with publicly accessible locations such as sidewalks and alleys yielding the highest arrest rates (31%–33%). In contrast, private settings like residences and apartments demonstrate significantly lower arrest outcomes, hovering around 10%. This disparity highlights a "visibility-enforcement gap," where open urban spaces facilitate proactive patrolling and immediate on-site intervention, while private domains remain challenging enforcement zones due to reporting delays and the absence of constant surveillance.

These findings suggest that law enforcement effectiveness is heavily dictated by environmental and procedural constraints, such as the legal requirements for warrants and the "reporting lag" inherent in residential crimes.

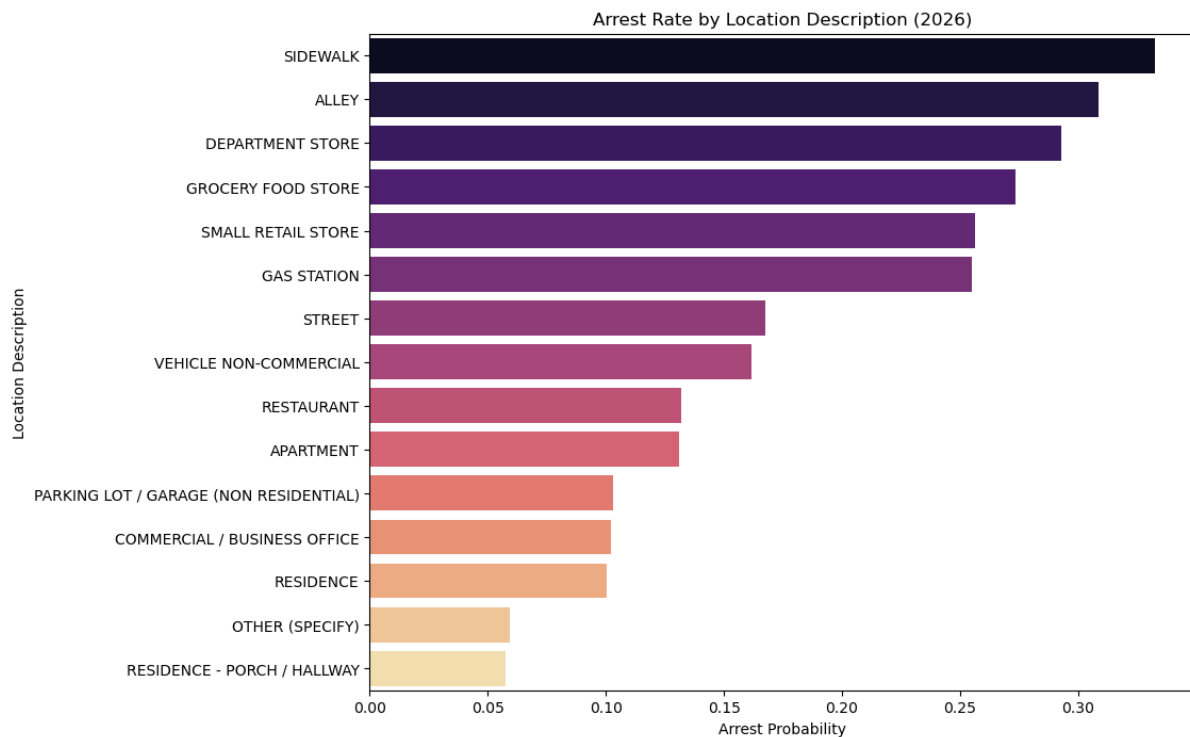


Figure2.Arrest Rate by Location Description

4. Interactive Chart and GitHub(link)

[IT5006 Phase 1 – Chicago Crime Dashboard · Streamlit](#)

The dashboard enables users to explore temporal and spatial crime patterns in Chicago from 2015 to 2024 through interactive filters, including year range and crime type selection.

Key visual components include a temporal line chart illustrating yearly crime trends, a spatial map visualizing geographic crime distributions, and a dynamic Top-5 crime category summary.

GitHub Link: [clear-train/Team8_IT5006_Predictive_Policing_AY2526Sem2](#)

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